

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Digital Signal Processing	Course Code:	EE3031
	Program:	BS Electrical Engineering	Semester:	Spring 2025
	Assignment Submission Date:	13-May 2025	Total Marks:	42
	Section:		Weight:	3
	Exam Type:	Assignment- 3	Page(s):	1
			CLO #	5

Student Name: _____ **Roll No.** _____ **Section:** _____

Instructions: 1. Submission instructions depending on whether it is a hardcopy/softcopy submission
2.

Question No. 1 (CLO No. 5)

Marks: 8

Design an FIR linear phase, digital filter approximating the ideal frequency response

$$H_d(\omega) = \begin{cases} 1, & \text{for } |\omega| \leq \frac{\pi}{6} \\ 0, & \text{for } \frac{\pi}{6} < |\omega| \leq \pi \end{cases}$$

- Determine the coefficients of a 7-tap filter based on the window method with a rectangular window.
- Determine and plot the magnitude and phase response of the filter.
- Repeat parts (a) and (b) using the Hamming window.
- Repeat parts (a) and (b) using a Bartlett window.

Question No. 2 (CLO No. 5)

Marks: 8

Repeat Question no. 1 for a bandstop filter having the ideal response

$$H_d(\omega) = \begin{cases} 1 & \text{for } |\omega| \leq \frac{\pi}{6} \\ 0 & \text{for } \frac{\pi}{6} < |\omega| \leq \frac{\pi}{3} \\ 1 & \text{for } \frac{\pi}{3} \leq |\omega| \leq \pi \end{cases}$$

Question No. 3 (CLO No. 5)

Marks: 16

Redesign the filter of Question no. 1 using the Hanning and Blackman windows.

Question No. 4 (CLO No. 5)

Marks: 10

A digital low-pass filter is required to meet the following specifications:

- Passband ripple: ≤ 1 dB
- Passband edge: 4 kHz
- Stopband attenuation: ≥ 40 dB
- Stopband edge: 6 kHz
- Sample rate: 24 kHz

The filter is to be designed by performing a bilinear transformation on an analog system function. Determine what order Butterworth, Chebyshev, and elliptic analog designs must be used to meet the specifications in the digital implementation.